Task 3: Linear Regression - Screenshot Documentation

1. Dataset Information

```
Dataset Shape: (1338, 7)

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   gender    1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

2. Data Preprocessing

	0
age	0
gender	0
bmi	0
children	0
smoker	0
region	0
charges	0

dtype: int64

```
np.int64(1)
```

```
Dataset Shape After Removing Duplicates: (1337, 7)
Duplicate Records: 0
```

3. Encoding Categorical Variables

	age	bmi	children	charges	gender_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.900	0	16884.92400	0	1	0	0	1
1	18	33.770	1	1725.55230	1	0	0	1	0
2	28	33.000	3	4449.46200	1	0	0	1	0
3	33	22.705	0	21984.47061	1	0	1	0	0
4	32	28.880	0	3866.85520	1	0	1	0	0

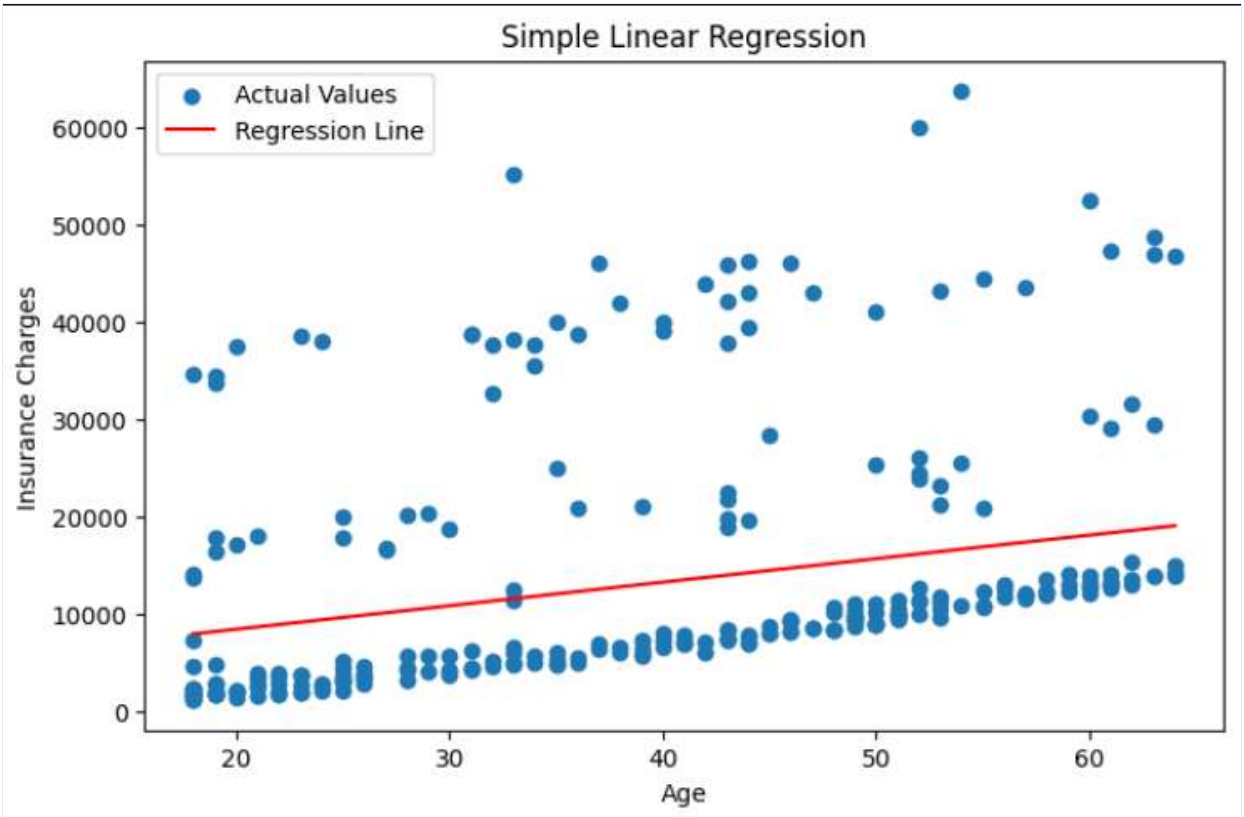
4. Splitting the Dataset

```
Training Data Shape: (1069, 8)
Testing Data Shape: (268, 8)
```

5. Simple Linear Regression

```
Coefficient: 242.25815190122503
Intercept: 3534.0917330710636
```

6. Simple Linear Regression Visualization



7. Multiple Linear Regression

```
Intercept: -11092.65229594595

Coefficients:
age : 248.21072022470446
bmi : 318.70144095235077
children : 533.009988798697
gender_male : -101.54205399029631
smoker_yes : 23077.764592868083
region_northwest : -391.7614547836092
region_southeast : -838.9196157269652
region_southwest : -659.1397515529586
```

8. Model Evaluation

```
Mean Absolute Error (MAE): 4177.045561036319
Mean Squared Error (MSE): 35478020.67523561
R2 Score: 0.8069287081198011
```

9. Actual vs Predicted Values

